

Simplify R:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Theta_{\gamma\gamma}^A \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^B \cdot \Theta_{z\beta}^A \\ -30 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A + 6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^B \cdot \Theta_{\alpha\beta}^A \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{\gamma\gamma}^A \\ -30 \cdot R_z^4 \cdot \Theta_{z\beta}^B \cdot \Theta_{z\beta}^A - 30 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A + 3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^B \cdot \Theta_{\alpha\beta}^A \end{array} \right] \quad (3b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{\alpha\alpha}^A \\ -30 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A - 30 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A + 3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^B \cdot \Theta_{\alpha\beta}^A \end{array} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{\alpha\alpha}^A \\ -60 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A - 15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A + 6 \cdot R_z^4 \cdot \Theta_{\alpha\beta}^B \cdot \Theta_{\alpha\beta}^A \end{array} \right] \quad (3d)$$

Simplify Quadrupoles:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +105 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A - 15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{\alpha\alpha}^A \\ -60 \cdot R_z^4 \cdot \Theta_{z\alpha}^B \cdot \Theta_{z\alpha}^A - 15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A + 6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\alpha\alpha}^A \end{array} \right] \quad (4a)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +45 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ -15 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{\alpha\alpha}^A \\ -15 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{zz}^A \\ +3 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\beta\beta}^A \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\alpha\alpha}^A \end{array} \right] \quad (4b)$$

Exploit Traceless Conditions:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +45 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ +6 \cdot R_z^4 \cdot \Theta_{\alpha\alpha}^B \cdot \Theta_{\alpha\alpha}^A \end{array} \right] \quad (5)$$

Multiply Out Sum:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +45 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ +6 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ +6 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \\ +6 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \end{array} \right] \quad (6a)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot \frac{1}{9} \cdot \left[\begin{array}{c} +51 \cdot R_z^4 \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ +6 \cdot R_z^4 \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ +6 \cdot R_z^4 \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{array} \right] \quad (6b)$$

Combining everything:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = \left[\begin{array}{c} +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{zz}^A \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{array} \right] \quad (7)$$

A:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = \begin{bmatrix} -\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{xx}^A \\ -\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{zz}^B \cdot \Theta_{yy}^A \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (8)$$

B:

$$+\frac{1}{9} \cdot T_{\alpha\beta\gamma\delta} \Theta_{\alpha\beta}^B \Theta_{\gamma\delta}^A = \begin{bmatrix} +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + \frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A \\ +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A + \frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \\ +\frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A + \frac{2}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (9a)$$

$$\xRightarrow{\text{added up}} \begin{bmatrix} +\frac{19}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{xx}^A \\ +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{xx}^A \\ +\frac{17}{3} \cdot R_z^{-5} \cdot \Theta_{xx}^B \cdot \Theta_{yy}^A \\ +\frac{19}{3} \cdot R_z^{-5} \cdot \Theta_{yy}^B \cdot \Theta_{yy}^A \end{bmatrix} \quad (9b)$$